Merge Sort

```
#include <bits/stdc++.h>
using namespace std;
#define max 10

int a[11] = {11, 45, 32, 45, 3, 67, 54, 345, 45, 43, 44};
int b[10];

void merging(int low, int mid, int high)
{
   int l1, l2, i;
   for (l1 = low, l2 = mid + 1, i = low; l1 <= mid && l2 <= high; i++)
   {
      if (a[l1] <= a[l2])
         b[i] = a[l1++];
      else
         b[i] = a[l2++];
   }
   while (l1 <= mid)
      b[i++] = a[l1++];
   while (l2 <= high)
      b[i++] = a[l2++];
   for (i = low; i <= high; i++)
      a[i] = b[i];
}

void sort(int low, int high)
{
   int mid;
   if (low < high)
   {
      mid = (low + high) / 2;
      sort(low, mid);
      sort(mid + 1, high);
```

```
        merging(low, mid, high);
    }
    else
    {
        return;
    }
}

int main()
{
    cout << "Array before sorting: " << endl;
    for (int i = 0; i <= max; i++)
    {
        cout << a[i] << " ";
    }
    sort(0, max);
    cout << "Array after sorting: " << endl;
    for (int i = 0; i <= max; i++)
    {
        cout << a[i] << " ";
    }
    return 0;
}
```

Output:

```
PS C:\Users\swarn\Repositories\C++> G++ .\mergeSort.cpp
PS C:\Users\swarn\Repositories\C++> ./a.exe
Array before sorting:
11 45 32 45 3 67 54 345 45 43 44 Array after sorting:
3 11 32 43 44 45 45 45 54 67 345
```